A4 scalar constructive measure: scoped T6 enactment

TECT verification-first repository · claim A4-SCALAR-SPECTRAL-CONSTRUCTIVE-MEASURE

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1. Purpose and scope

This addendum records the tier review and independent operator reproduction for the theorem proved in `a4-scalar-spectral-constructive-measure-260718-v1.0`. It does not replace or shorten that proof. Its only mathematical action is to enact the pre-registered T5-to-T6 transition after a fresh operator-side execution.

The theorem remains the finite-volume real-scalar statement on the fixed three-torus with sharp max-norm spectral projectors:

$$K(k) = m_{\mathrm{sh}}^2 + Y(|k|^2 - q_0^2)^2, \qquad V(\phi) = \int_{\mathbb{T}^3} \left(\frac{\lambda}{4}\phi^4 + \frac{\gamma}{6}\phi^6\right) dx. \tag{1.1}$$

The named hypotheses are

$$\begin{aligned} \texttt{A1-SHELL-POSITIVITY}: &\quad m_{\mathrm{sh}}^2 > 0, \quad Y > 0, \quad q_0 \geq 0, \\ \texttt{A2-H2-SEXTIC-COERCIVITY}: &\quad \lambda \in \mathbb{R}, \quad \gamma > 0. \end{aligned} \tag{1.2}$$

The hard dependency `A1-SCALAR-ANALYTIC-BRANCH` is already T6. The other production manifests remain soft anchors and are not silently promoted to theorem inputs.

Under (1.2), the full Galerkin sequence converges weakly on real L^2 to the finite-volume Gibbs measure. The common-Gaussian lifted densities converge in L^1 and total variation, and finite-degree smeared cylinder-polynomial correlations converge. The proof uses the trace-class q^{-4} covariance, the direct Gaussian L^6 tail, the exact quartic-sextic lower bound, uniform partition bounds, and common-space density convergence.

1 Independent operator reproduction

Jusang Lee executed the one-command verifier from the clean checkout at commit

`7eee2fe84887cc4ccdd311c75095ec84bc9d0d45`

and wrote a separate immutable result at

```
claims/A4-SCALAR-SPECTRAL-CONSTRUCTIVE-MEASURE/runs/
2026-07-18-jusang-independent/result.json
```

The artefact SHA-256 is

`fb2d81ba75d95972e55c675146fbf84cb8209847ea031abdaa0a5f44b111935c`

and records Python 3.12.10 on Windows 11. It reports

```
PASS: primary (17/17)
PASS: independent (14/14)
ASSERTS: 31/31
A4-SCALAR-CONSTRUCTIVE-INTEGRATED-PASS
```

with zero failures, zero failed source-hash rows, zero failed audit rows, and return code zero for both executable routes.

The operator artefact has the same content hash as the accepted internal integrated result. This equality is expected for a deterministic verifier and is not used as the evidence of execution independence. Independence is recorded by the separate operator-created path and filesystem event, the clean source commit, and the fresh subprocess return codes. Algorithmic diversity is supplied separately by the non-importing scalar-loop reconstruction, which does not import the vectorized primary audit.

2 Tier decision

The T6 requirements are now met:

1. The v1.0 note is a self-contained analytic proof under the two named hypotheses in (1.2).
2. The primary audit derives the trace, stability, partition, and Gaussian tail bounds and passes 17/17.
3. A non-importing implementation independently reconstructs the shell count, scalar Fourier traces, potential minimum, Gaussian moments, and partition bounds and passes 14/14.
4. The integrated verifier hash-binds all three sources and passes 31/31 in both the internal and operator-side executions.
5. The claim card, manifest, proof PDF, evidence JSONs, falsifier, and no-overclaim boundary form an available one-command reproduction package.

Therefore the claim is enacted as

$$\boxed{\text{T6 CONDITIONAL-THEOREM@FINITE-VOLUME-REAL-SCALAR-SPECTRAL}}. \tag{3.1}$$

This is a T5-to-T6 promotion, not a T7 claim. T7 is prohibited because the result is deliberately conditional and scoped, has no public standalone reproduction bundle, and does not cover the full physical-domain functional.

3 Quantitative sanity record

The independent scalar-loop covariance trace agrees with the primary vectorized trace to at most 1.32×10^{-16} relative error at cutoff 8. The production-local shell mass reconstructs as 0.260000000009475, agreeing with the stored 0.26 within the declared source precision. Conservative full covariance-trace upper bounds are 3029.399163006103 and 596.857766699839 for the two audit anchors. The exact pinned-coefficient stability constant is 0.0025246087994716246 per unit volume. These are derived reproduction checks, not physical predictions.

4 Devil's-advocate review

1. **Identical internal and operator JSON hashes prove that the operator file was copied. DISMISSED.** The verifier is deterministic; the independent path, creation event, clean source commit, and fresh subprocess return codes record the separate execution.
2. **Operator reproduction merely repeats one implementation. VALID-with-mitigation.** The one command launches both the vectorized primary route and a non-importing scalar-loop reconstruction. External mathematical peer review remains desirable but is not mislabeled as completed.

3. **The T6 claim inherits a sub-T6 hard dependency. DISMISSED.** The sole hard dependency is T6; the positivity and coercivity inputs are explicit named hypotheses, and production manifests are soft anchors only.

4. **Scalar T6 closes the derivative three-component Class-II measure. UPHELD as false.** Rough Gaussian samples do not automatically define derivative-current squares; that extension is excluded.

5. **Finite-volume regulator removal proves a phase transition. UPHELD as false.** No thermodynamic limit or spontaneous symmetry breaking is proved.

6. **This promotion closes Sector A or BCC selection. UPHELD as false.** P5 synthesis remains a separate package and BCC existence or selection remains explicitly outside it.

5 Falsifier and next boundary

The T6 status is revoked if the source or evidence hashes drift; either route fails; the trace, Gaussian L^6, stability, Jensen, density-convergence, or correlation argument admits a counterexample in the declared scope; or the operator artefact is shown not to be a fresh execution. Such a finding reopens `A4-CONSTRUCTIVE-MEASURE-CLOSURE`.

The next Sector-A action is P5 synthesis. P5 may cite this theorem only at its finite-volume real-scalar spectral scope. It must not silently extend it to the derivative Class-II measure, unsmeared composite operators, infinite volume, phase transition, finite-difference Route B, or BCC existence and selection.

Result footer

```
Result ID:               A4-SCALAR-SPECTRAL-CONSTRUCTIVE-MEASURE
Precise statement:       Full spectral Galerkin Gibbs sequence converges
                         weakly on L2; lifted densities converge in L1/TV;
                         smeared cylinder-polynomial correlations converge.
Scope:                   T6 CONDITIONAL-THEOREM@FINITE-VOLUME-REAL-SCALAR-
                         SPECTRAL; periods 16, m_sh^2>0, Y>0, q0>=0,
                         lambda real, gamma>0, beta=1.
Dependencies:            A1-SCALAR-ANALYTIC-BRANCH; named hypotheses
                         A1-SHELL-POSITIVITY and A2-H2-SEXTIC-COERCIVITY.
Evidence grade:          ANALYTIC + EXACT + EXECUTED + CONDITIONAL.
Reproduction command:    python codes/foundations/
                         a4_scalar_constructive_measure_verify.py
Expected output:         primary 17/17; independent 14/14; ASSERTS 31/31;
                         A4-SCALAR-CONSTRUCTIVE-INTEGRATED-PASS.
Operator evidence:       2026-07-18-jusang-independent/result.json;
                         SHA-256 fb2d81ba75d95972e55c675146fbf84c...
Falsification gate:      Failed proof step, route, source/evidence hash, or
                         operator-execution provenance.
Tier before / after:     T5 / T6 after independent operator reproduction.
No-overclaim statement:  Not derivative Class-II, unsmeared composites,
                         infinite volume, phase transition, Route B, BCC,
                         T7, or P5 synthesis.
Next required action:    Build the separate P5 Sector-A synthesis and
                         reproduction bundle with these exclusions intact.
```